CA3 - Classification of Wine

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Use machine learning to develop a system/classifier capable of distinguishing between high-quality and low-quality wines based on measurable characteristics.

Your task is to create a classifier that uses the available wine characteristics to sort the wines into the correct categories.

The dataset contains several physicochemical properties of wine that can be used to predict its quality. These properties describe characteristics such as acidity, sugar content, density, sulphates, alcohol level, and other chemical measurements.

The success of your classification system depends on your ability to explore the data, identify useful patterns, and train a model that can accurately distinguish between good and bad wines.

Sample: one row, one wine

Feature: Attribute, variable, column (pH, alcohol)

Target: quality (0=bad, 1=good)

Feature matrix: X, all columns together (835 rows x 11 columns)

Target vector: y, the target column (835 values of 0 or 1)

Sources and acknowledgements

Material in this notebook that did not originate with us:

- The `evaluate` function and the tuning-loop pattern follow the approach shown in the lecture notebooks `03_svm_gridsearch_iris.ipynb` and `03_knn_hyperparameters_validation_set.ipynb`, and the procedure set out in `CA3_Workflow.pdf`.
- The code in section 5, section 8 and section 9, the docstring on `evaluate`, the axis labels and titles added to the plots in sections 3 and 6, and the accompanying `CA3_Group6.py` were drafted with assistance from an AI assistant (Claude, Anthropic) and then reviewed by us.

- Discussion of the results — how to compare models on the same splits, why the maximum over many settings is biased upward, and how to read the train/validation gap — also drew on that assistance.

The dataset is the wine quality data supplied with the assignment.

1. Imports

In [1]:
```
import numpy as np                  # numbers and arrays
import pandas as pd                 # tables (DataFrames)
import matplotlib.pyplot as plt     # basic plotting
import seaborn as sns               # nicer statistical plots on top of matpl
import sklearn as skl               # machine learning library

from sklearn.model_selection import train_test_split # splits data into trai
from sklearn.preprocessing import StandardScaler # rescales features to have
from sklearn.linear_model import Perceptron, LogisticRegression # linear cla
from sklearn.svm import SVC     # support vector machine, a more powerful li
from sklearn.tree import DecisionTreeClassifier # decision tree, a non-linea
from sklearn.ensemble import RandomForestClassifier     # random forest, a m
from sklearn.neighbors import KNeighborsClassifier  # k-nearest neighbors, a
```

2. Reading data

In [2]:
```
train = pd.read_csv("train.csv")                # 11 features + quality + l
test = pd.read_csv("test.csv")                  # 11 features + Id, no qual
sample = pd.read_csv("sample_submission.csv")   # the format Kaggle expects
print("train:", train.shape, " test:", test.shape, " sample:", sample.shape)

# The feature columns: everything except the target and the row label
features = [col for col in train.columns if col not in ("quality", "Id")]
print(len(features), "features:", features)
```

```
train: (835, 13)  test: (209, 12)  sample: (209, 2)
11 features: ['fixed acidity', 'volatile acidity', 'citric acid', 'residual
sugar', 'chlorides', 'free sulfur dioxide', 'total sulfur dioxide', 'densit
y', 'pH', 'sulphates', 'alcohol']
```

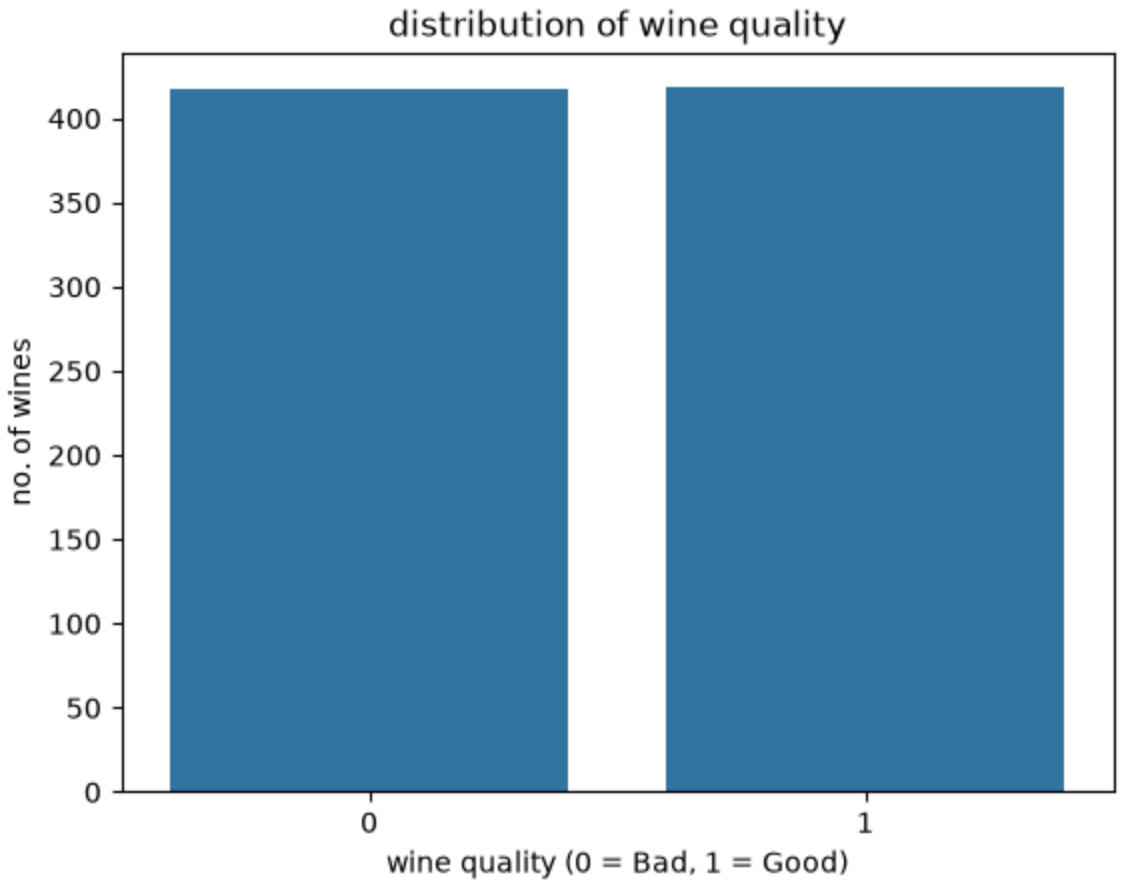
Train.csv has 835 rows and 13 columns

3. Data exploration and visualisation

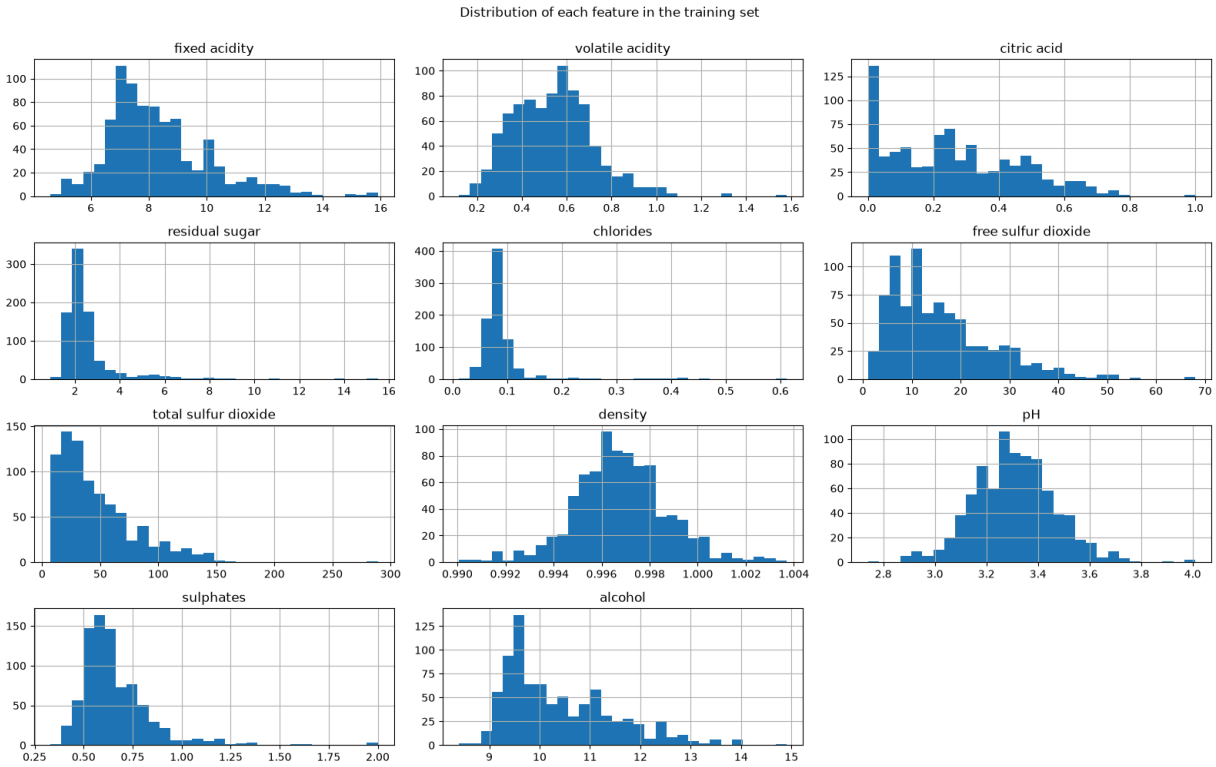
In [3]:
```
# Explore the training set
train.info()                        # shows entries, column types and memory
train.describe().T                  # .T turns the table: one row per feature
print(train["quality"].value_counts()) # count of each quality value
print("Missing values:", train.isnull().sum().sum()) # total number of missi
print("Duplicate rows (ignoring Id):", train.duplicated(subset=features + ["
```

```
<class 'pandas.DataFrame'>
RangeIndex: 835 entries, 0 to 834
Data columns (total 13 columns):
 #   Column                Non-Null Count  Dtype  
---  ------                --------------  -----  
 0   fixed acidity         835 non-null    float64
 1   volatile acidity      835 non-null    float64
 2   citric acid           835 non-null    float64
 3   residual sugar        835 non-null    float64
 4   chlorides             835 non-null    float64
 5   free sulfur dioxide   835 non-null    float64
 6   total sulfur dioxide  835 non-null    float64
 7   density               835 non-null    float64
 8   pH                    835 non-null    float64
 9   sulphates             835 non-null    float64
 10  alcohol               835 non-null    float64
 11  quality               835 non-null    int64  
 12  Id                    835 non-null    int64  
dtypes: float64(11), int64(2)
memory usage: 84.9 KB
quality
1    418
0    417
Name: count, dtype: int64
Missing values: 0
Duplicate rows (ignoring Id): 75
```

In [4]:
```
# distribution of wine quality
sns.countplot(data=train, x="quality")
plt.xlabel("wine quality (0 = Bad, 1 = Good)")
plt.ylabel("no. of wines")
plt.title("distribution of wine quality")
plt.show()
# Wine quality between good and bad is eqally distributed
```

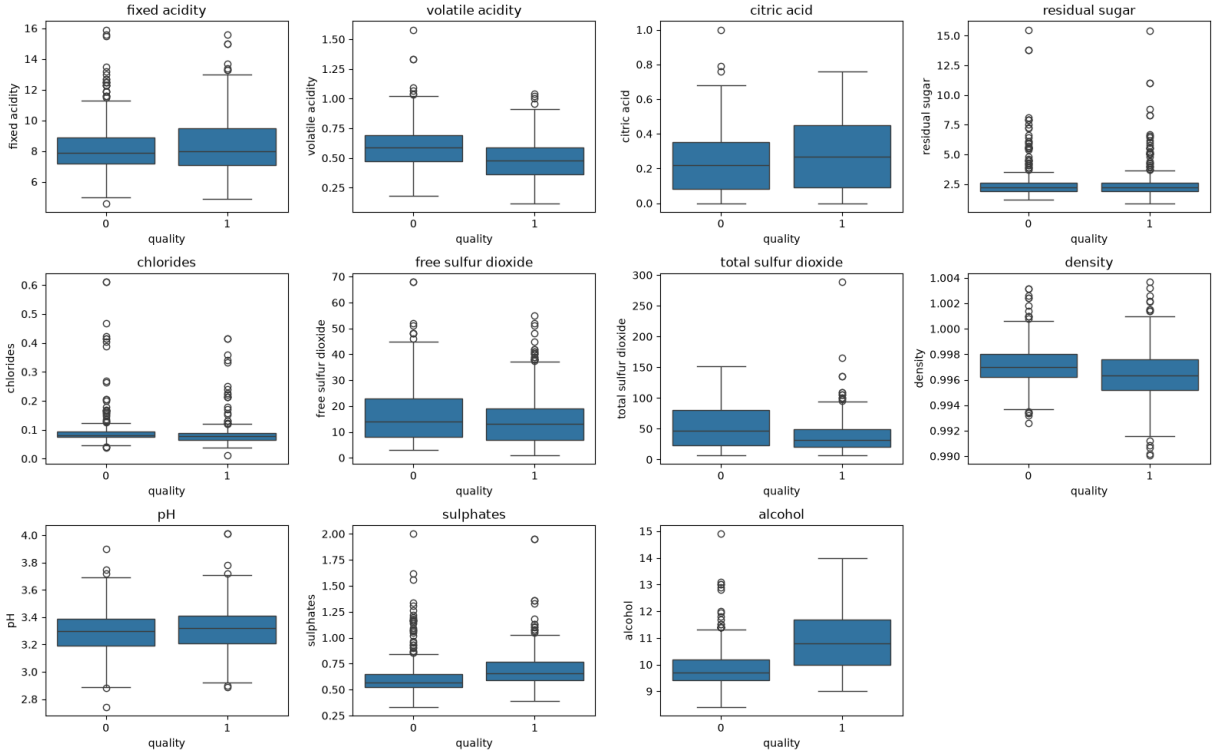


In [5]:
```
# Visualize the distribution of each feature in the training set
train[features].hist(figsize=(16, 10), bins=30)
plt.suptitle("Distribution of each feature in the training set", y=1.00)
plt.tight_layout()
plt.show()
# bell-shaped for pH and density
# right skewed for residual sugar, chlorides, total sulfur dioxide
```



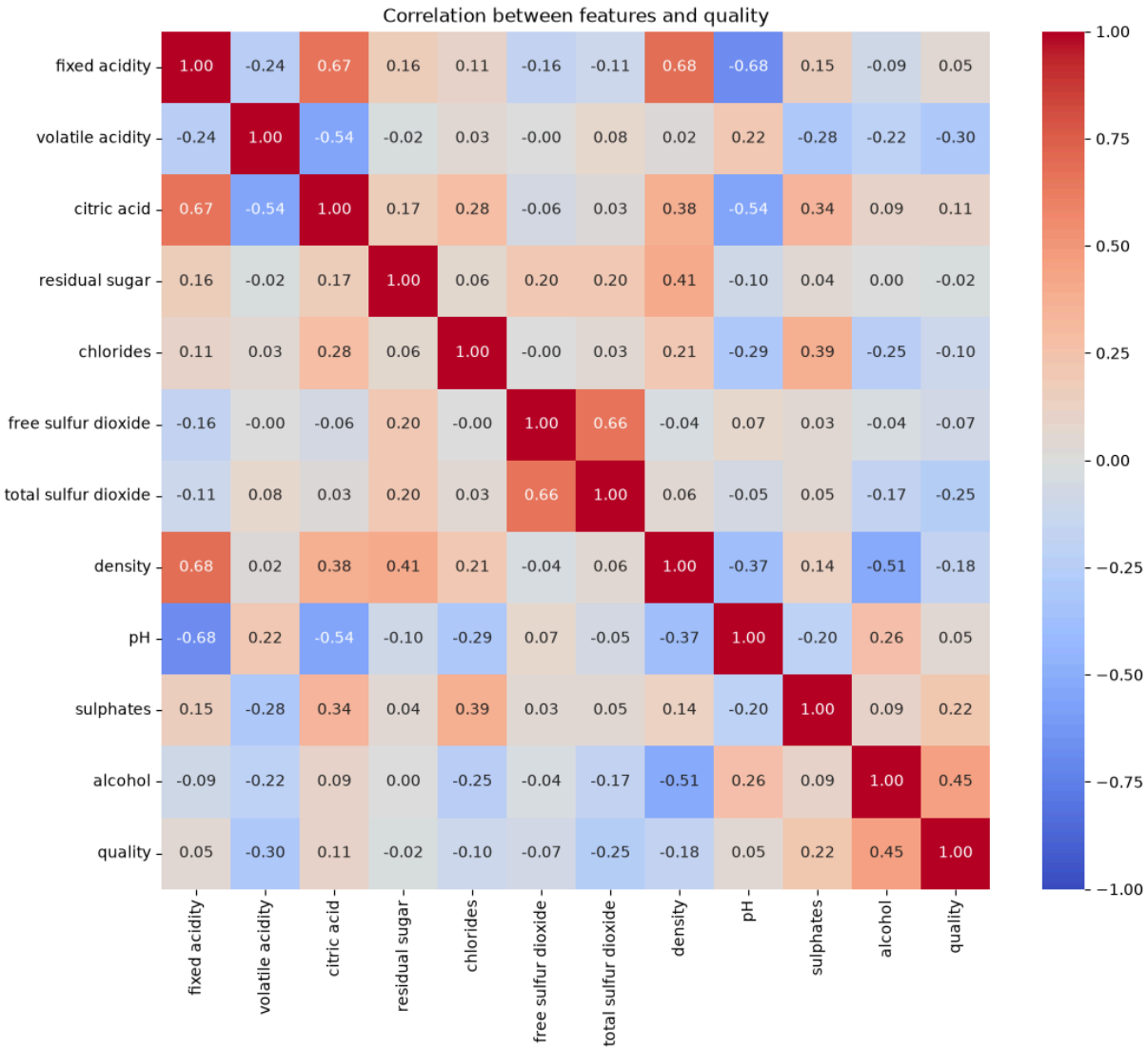
In [6]:
```
# Visualize the distribution of each feature in the training set, grouped by
fig, axes = plt.subplots(3, 4, figsize=(16, 10))     # 3 x 4 grid: 12 panels
for ax, col in zip(axes.flatten(), features):          # one panel per featu
    sns.boxplot(data=train, x="quality", y=col, ax=ax)
    ax.set_title(col)
axes.flatten()[-1].set_visible(False)                  # hide the empty 12th
plt.tight_layout()
plt.show()

# Good quality wine (1) has generally higher alcohol level
# Good quality wine usually has lower volatile acidity and total sulfur diox
# Outlires in several features (e.g. residual sugar and chlorides)
```



In [7]:
```
# Correlation matrix and heatmap
corr = train[features + ["quality"]].corr() # compute correlation matrix for
plt.figure(figsize=(12, 10)) # set fig size
sns.heatmap(corr, annot=True, # gives each cell a the numeric value
            fmt=".2f", # format numbers to 2 decimal places
            cmap="coolwarm",
            vmin=-1,
            vmax=1)
plt.title("Correlation between features and quality")
plt.show()

print(corr["quality"].drop("quality").sort_values())   # correlation of each
""
```




```
volatile acidity       -0.302238
total sulfur dioxide   -0.250267
density                -0.179263
chlorides              -0.103287
free sulfur dioxide    -0.074444
residual sugar         -0.020452
pH                      0.049578
fixed acidity           0.051339
citric acid             0.111017
sulphates               0.219372
alcohol                 0.451935
Name: quality, dtype: float64
```

Out[7]: ''

- Alcohol is the dominant single predictor at +0.452, so higher alcohol associates with good wine.
- Volatile acidity is the strongest negative at −0.302. It's the compound that makes wine taste of vinegar, so this assosiates with bad wine.
- No feature is strong enough to classify on its own. The largest is 0.452; a correlation that size explains roughly 20% of the variation, so this needs a model combining

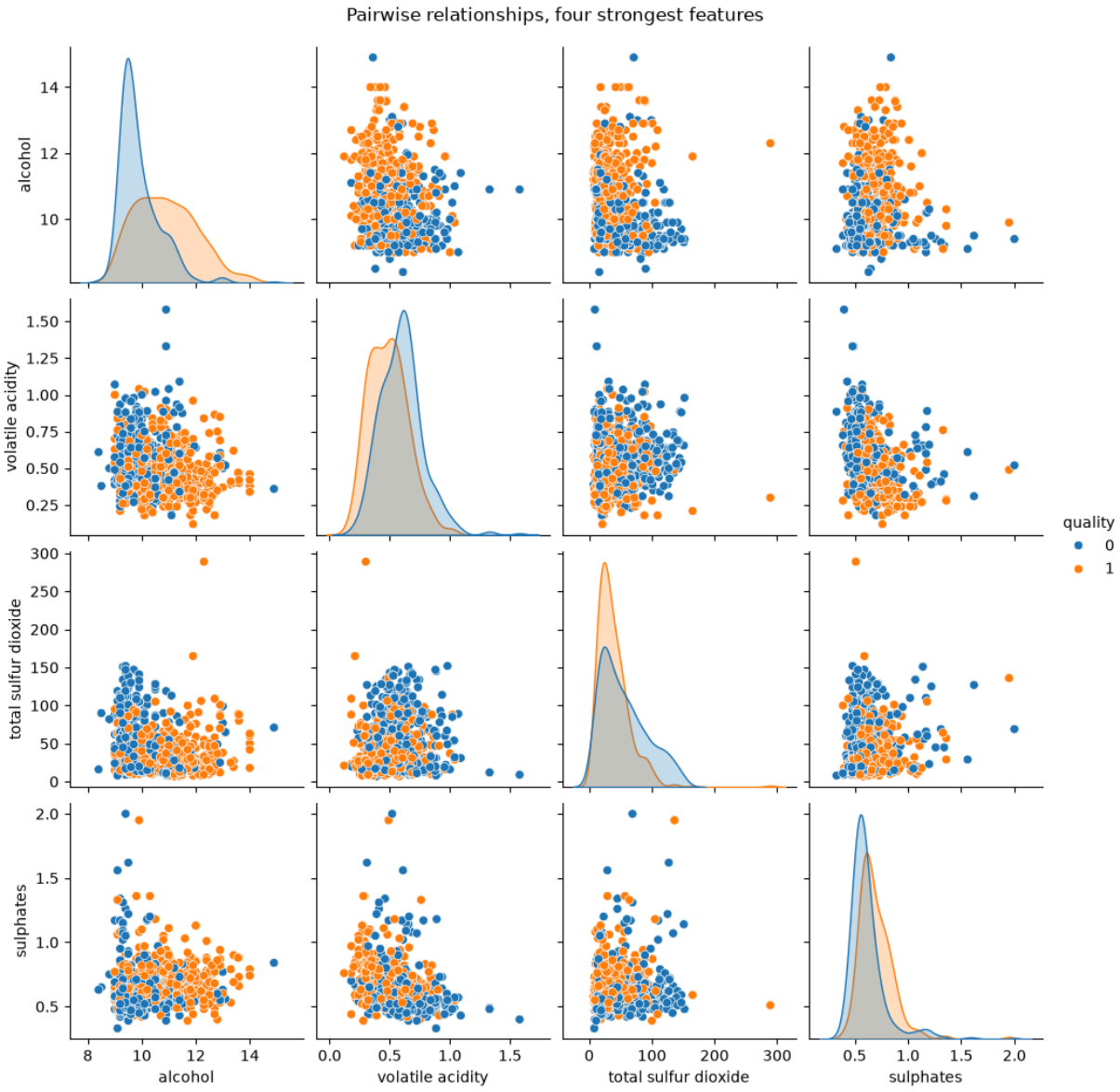
several features.

- All features and values ar valuable, because Pearson only detects linear relationships and a weak score here doesn't mean a feature is useless in combination.

In [8]:
```
# The 4 features most correlated with quality, visualized by their pairwise
top = corr["quality"].drop("quality").abs().sort_values(ascending=False).inc
    # take quality's correlations, drop the self-match, ignore the signs, ra

g = sns.pairplot(train, vars=top, hue="quality")
g.figure.suptitle("Pairwise relationships, four strongest features", y=1.02)
plt.show()

# The distributions of the two quality classes highly overlap
# although alcohol shows more separation
# combining multiple features may improve classification
```



4. Data cleaning

In [9]:
```
data = train.drop(columns=["Id"]) # drop the Id column, if it was still ther

# Check for duplicate rows with conflicting labels
duplicates = data.duplicated(subset=features, keep=False) # mark all rows th
conflicts = data[duplicates].groupby(features)["quality"].nunique() # count
print("Duplicate groups with conflicting labels:", (conflicts > 1).sum())

# Keep the first copy of every exact duplicate
data = data.drop_duplicates()
print("Rows after removing exact duplicates:", len(data))
```

```
Duplicate groups with conflicting labels: 0
Rows after removing exact duplicates: 760
```

In [10]:
```
print("Negative values:", (data[features] < 0).sum().sum()) # count of negat
print("pH outside 2.5 to 4.5:", ((data["pH"] < 2.5) | (data["pH"] > 4.5)).su
print("Alcohol outside 5 to 16 %:", ((data["alcohol"] < 5) | (data["alcohol"
```

```
Negative values: 0
pH outside 2.5 to 4.5: 0
Alcohol outside 5 to 16 %: 0
```

In [11]:
```
z = (data[features] - data[features].mean()) / data[features].std() # comput
print((z.abs() > 3).sum())                          # outliers per feature
outlier_rows = (z.abs() > 3).any(axis=1)            # True if ANY feature has
print("Rows with |z| > 3 in any feature:", outlier_rows.sum()) # count of ro
```

```
fixed acidity           7
volatile acidity        3
citric acid             1
residual sugar         15
chlorides              16
free sulfur dioxide     9
total sulfur dioxide    4
density                10
pH                      4
sulphates              12
alcohol                 6
dtype: int64
Rows with |z| > 3 in any feature: 68
```

5. Data exploration and visualisation after cleaning

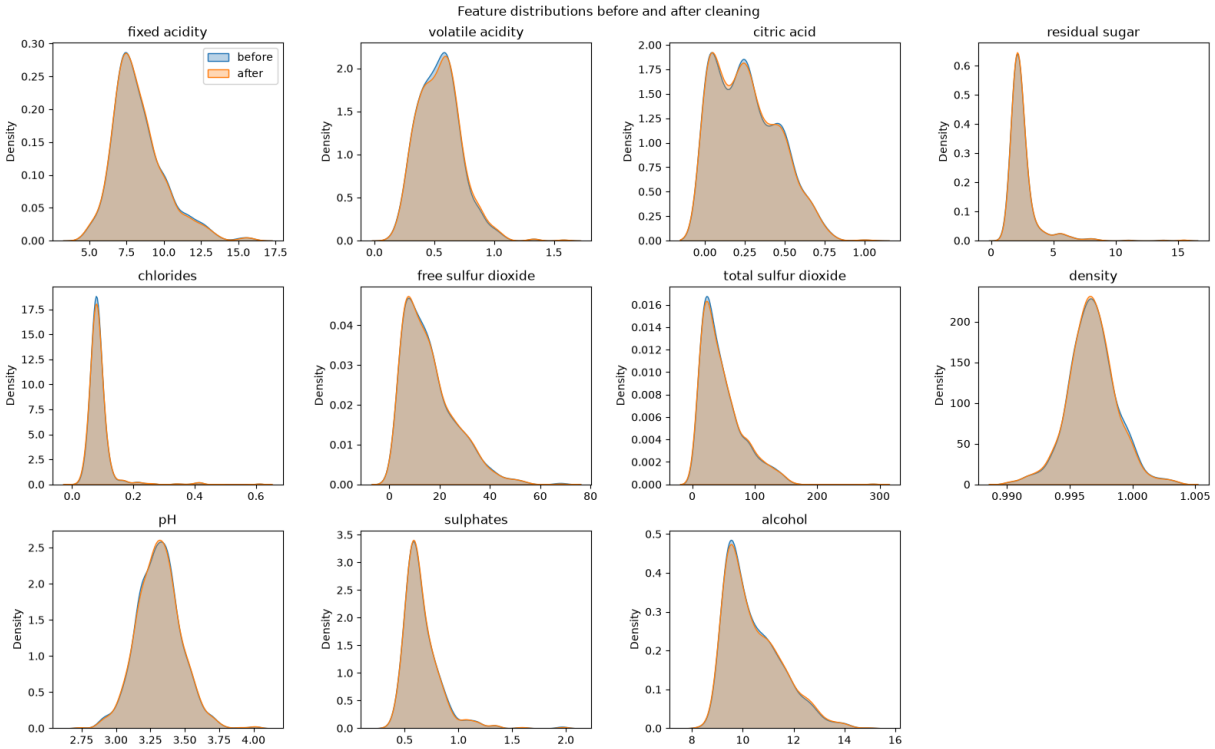
In [12]:
```
# How much did cleaning change the data?
print("Rows before:", len(train), " after:", len(data))
print(data["quality"].value_counts(normalize=True).round(3))

# The same feature before and after, drawn on one pair of axes
fig, axes = plt.subplots(3, 4, figsize=(16, 10))
for ax, col in zip(axes.flatten(), features):
    sns.kdeplot(train[col], ax=ax, label="before", fill=True, alpha=.3)
    sns.kdeplot(data[col],  ax=ax, label="after",  fill=True, alpha=.3)
    ax.set_title(col)
```

```
    ax.set_xlabel("")
axes.flatten()[-1].set_visible(False)   # hide the empty 12th panel
axes.flatten()[0].legend()
fig.suptitle("Feature distributions before and after cleaning")
plt.tight_layout()
plt.show()

# Did any correlation with quality move?
before = train[features + ["quality"]].corr()["quality"].drop("quality")
after  = data[features + ["quality"]].corr()["quality"].drop("quality")
comparison = pd.DataFrame({"before": before, "after": after})
comparison["change"] = comparison["after"] - comparison["before"]
print(comparison.round(3).sort_values("after"))
```

```
Rows before: 835  after: 760
quality
0    0.501
1    0.499
Name: proportion, dtype: float64
```




```
                      before  after  change
volatile acidity      -0.302 -0.299   0.004
total sulfur dioxide  -0.250 -0.262  -0.012
density               -0.179 -0.179  -0.000
chlorides             -0.103 -0.106  -0.003
free sulfur dioxide   -0.074 -0.089  -0.015
residual sugar        -0.020 -0.016   0.005
pH                     0.050  0.042  -0.008
fixed acidity          0.051  0.055   0.003
citric acid            0.111  0.118   0.007
sulphates              0.219  0.209  -0.010
alcohol                0.452  0.451  -0.001
```

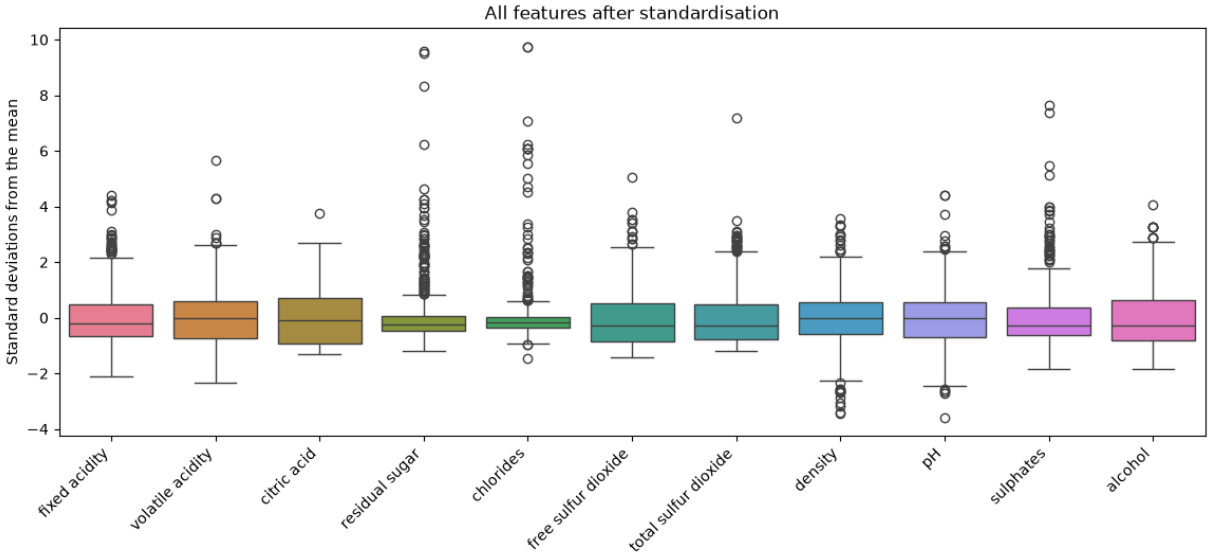
Distribution: the balance was 50.1/49.9 before and 50.1/49.9 after. The duplicates that came out were split evenly between the classes.

6. Data preprocessing

In [13]:
```
# Makning the feature matrix and target vector for training the model

X = data[features].to_numpy() # one row per wine, one column per feature (76
y = data["quality"].to_numpy() # one row per wine, one column per target var
```

In [14]:
```
scaled = pd.DataFrame(StandardScaler().fit_transform(X), columns=features) #
plt.figure(figsize=(14, 5))
sns.boxplot(data=scaled)
plt.xticks(rotation=45, ha="right")
plt.title("All features after standardisation")
plt.ylabel("Standard deviations from the mean")
plt.show()
```



In [15]:
```
def evaluate(make_model, X, y, n_splits=50, scale=True, drop_outliers=False)
    """Score a model by averaging over many random train/validation splits.

    The course rules forbid GridSearchCV and cross_val_score, so this is our
    We repeat the split n_splits times with a different random_state each pa
    Lecture 12 introduces this as the stand-in for k-fold cross-validation.

    Because random_state counts 0, 1, 2, ..., every model passed to this
    function sees the identical set of splits. That is what makes comparing
    different classifiers in section 7 a fair test.

    Parameters
    ----------
    make_model : callable
        A no-argument function that RETURNS a new, unfitted model, for examp
        lambda: LogisticRegression(C=1, max_iter=1000). It has to be a funct
        rather than a model object so that every split gets a fresh estimato
        with no weights left over from the previous pass.
    X : ndarray, shape (n_samples, n_features)
        The cleaned feature matrix.
    y : ndarray, shape (n_samples,)
```

```
    The 0/1 quality labels.
n_splits : int, default 50
    How many random splits to average over. Drop to about 10 while
    exploring a large grid, then raise it back for the numbers reported
scale : bool, default True
    Standardise the features. True for distance- and gradient-based mode
    (KNN, SVC, logistic regression, perceptron). False for decision tree
    and random forests, which split one feature at a time on a threshold
    and are therefore unaffected by rescaling.
drop_outliers : bool, default False
    Remove training rows where any feature lies more than 3 standard
    deviations from its mean. Whether this helps is an empirical questio
    so it is an option to measure rather than something applied by defau

Returns
-------
(float, float, float)
    Mean training accuracy, mean validation accuracy, and the standard
    deviation of validation accuracy across the splits. The gap between
    first two shows overfitting; the third shows how stable the model is
"""
train_scores, val_scores = [], []
for randome_split in range(n_splits):
    # 1. A new random split, the same one for every model
    # A different split each pass, but a reproducible one, since randome
    # stratify=y keeps the 50/50 class balance in both halves
    X_tr, X_va, y_tr, y_va = train_test_split(
        X, y, test_size=0.3, random_state=randome_split, stratify=y)

    # 2. Remove outliers from the training part only makes the test easi
    if drop_outliers:
        z_tr = (X_tr - X_tr.mean(axis=0)) / X_tr.std(axis=0)
        keep = (np.abs(z_tr) < 3).all(axis=1)
        X_tr, y_tr = X_tr[keep], y_tr[keep]

    # 3. Scale with a scaler fitted on the training part only
    # fit_transform learns the mean and standard deviation; transform on
    # the full data instead would let validation rows influence the trai
    # That is data leakage: it runs without any error and quietly makes
    if scale:
        sc = StandardScaler()
        X_tr = sc.fit_transform(X_tr)
        X_va = sc.transform(X_va)


    # 4. A fresh model, trained and scored
    # A brand new model every pass. make_model() builds it here
    # so no weights survive from the previous split.
    model = make_model()
    model.fit(X_tr, y_tr)
    train_scores.append(model.score(X_tr, y_tr))
    val_scores.append(model.score(X_va, y_va))

# np.std measures spread across the splits: a large value means the mode
# accuracy depends heavily on which rows it happened to train on.
return np.mean(train_scores), np.mean(val_scores), np.std(val_scores)
```

```
tr, va, sd = evaluate(lambda: LogisticRegression(max_iter=1000), X, y)
print(f"train {tr:.3f}, validation {va:.3f} ± {sd:.3f}")
```

```
train 0.750, validation 0.743 ± 0.028
```

7. Evaluating different models and hyperparameters (using train/test splits)

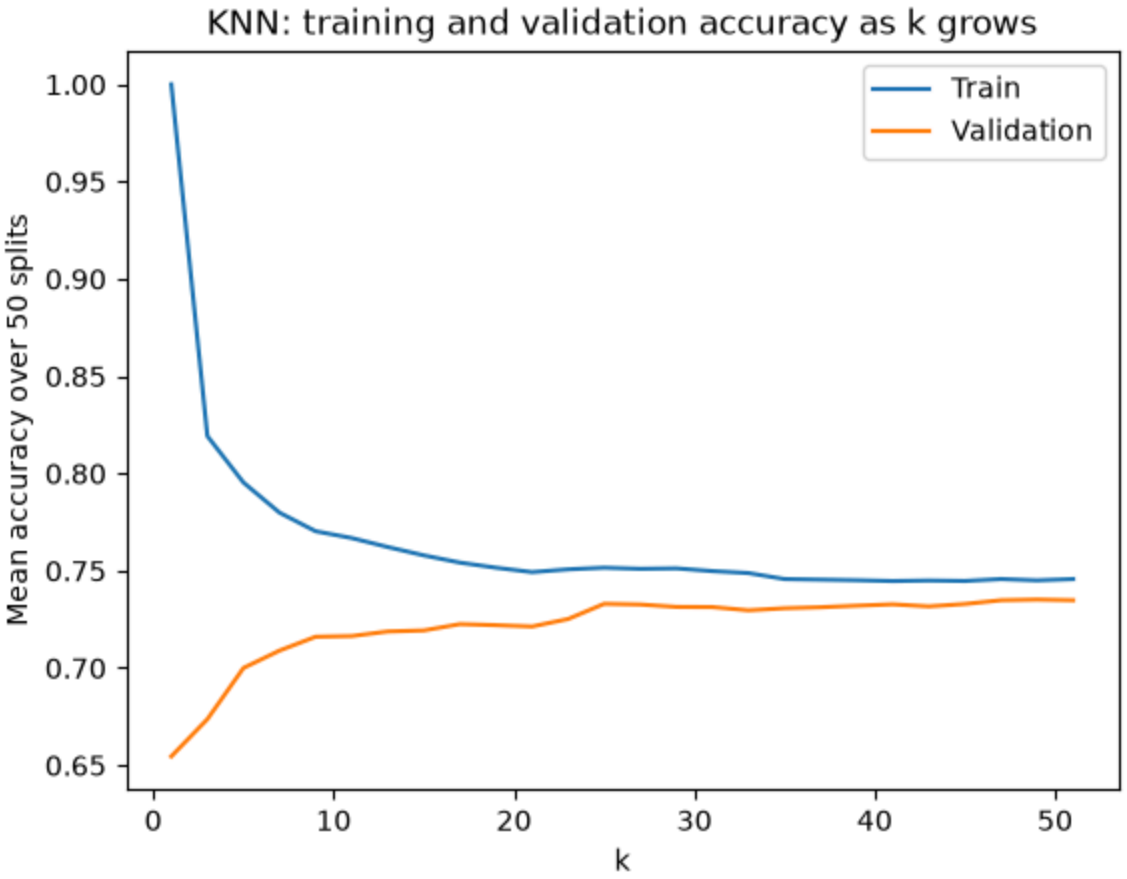
In [16]:
```
results = []   # run ONCE; every tuning cell appends to this
```

In [17]:
```
# Tune the learning rate (eta0) for the Perceptron
for eta in [0.0001, 0.001, 0.01, 0.1, 1.0]:
    tr, va, sd = evaluate(lambda: Perceptron(eta0=eta, max_iter=100, random_
    results.append({"model": "Perceptron", "settings": f"eta0={eta}",
                    "train": tr, "val": va, "std": sd})
```

In [18]:
```
# Tune the regularization strength (C) for the Logistic Regression
for c in [0.001, 0.01, 0.1, 1, 10, 100, 1000]:
    tr, va, sd = evaluate(lambda: LogisticRegression(C=c, max_iter=1000), X,
    results.append({"model": "Logistic regression", "settings": f"C={c}",
                    "train": tr, "val": va, "std": sd})
```

In [19]:
```
# Tune the number of neighbors (k) and distance metric (p) for the KNN
for k in range(1, 52, 2):          # odd k from 1 to 51
    for p in [1, 2]:               # 1 = Manhattan, 2 = Euclidean
        tr, va, sd = evaluate(lambda: KNeighborsClassifier(n_neighbors=k, p=
        results.append({"model": "KNN", "settings": f"k={k}, p={p}", "k": k,
                        "train": tr, "val": va, "std": sd})

# Training vs validation accuracy as k grows (Euclidean distance)
res = pd.DataFrame(results)
knn_p2 = res[(res["model"] == "KNN") & (res["p"] == 2)]
plt.plot(knn_p2["k"], knn_p2["train"], label="Train")
plt.plot(knn_p2["k"], knn_p2["val"], label="Validation")
plt.xlabel("k")
plt.ylabel("Mean accuracy over 50 splits")
plt.title("KNN: training and validation accuracy as k grows")
plt.legend()
plt.show()
```



In [20]:
```
# DecisionTree
from sklearn.tree import DecisionTreeClassifier

# evaluating different maximum tree depths using the repeated train/validati
for depth in [2, 3, 4, 5, 6, 8, 10, None]:

    tr, va, sd = evaluate(
        lambda: DecisionTreeClassifier(
            criterion='gini',
            max_depth=depth,
            random_state=1
        ),
        X, y,
        scale=False
    )
# saving the scores
    results.append({
        "model": "Decision Tree",
        "settings": f"max_depth={depth}",
        "train": tr,
        "val": va,
        "std": sd
    })
```

In [21]:
```
# Random forest

from sklearn.ensemble import RandomForestClassifier
```

```
# looping through combinations of n_estimators and max_depth and evaluating
for depth in [3, 5, 10, None]:
    for n in [50, 100, 200]:

        tr, va, sd = evaluate(
            lambda: RandomForestClassifier(
                n_estimators=n,
                criterion='gini',
                max_depth=depth,
                random_state=1,
                n_jobs=-1
            ),
            X, y,
            scale=False
        )
# saving the scores
        results.append({
            "model": "Random Forest",
            "settings": f"n_estimators={n}, max_depth={depth}",
            "train": tr,
            "val": va,
            "std": sd
        })
```

In [22]:

```
# SVC
from sklearn.svm import SVC

# Test linear and rbf kernels, each with 5 different regularization (c) valu
for kernel in ['linear', 'rbf']:
    for c in [0.01, 0.1, 1, 10, 100]:

        tr, va, sd = evaluate(
            lambda: SVC(
                kernel=kernel,
                C=c,
                gamma='scale',
                random_state=1
            ),
            X, y
        )
# saving the scores
        results.append({
            "model": "SVM",
            "settings": f"kernel={kernel}, C={c}",
            "train": tr,
            "val": va,
            "std": sd
        })
```

In [23]:

```
# Comparing the 6 classifiers

results_df = pd.DataFrame(results)

best_models = (
    results_df
```

```
    .sort_values("val", ascending=False)
    .drop_duplicates(subset="model")
    .sort_values("val", ascending=False)
)

print(best_models.to_string(index=False))

# Random forest is currently the best classifier with 0.755614 accurasy, but
# Logistic regression has the least val and train gap (about 1%) which sugge
# Perceptron has the lowest validation accuracy (0.663158) and the biggest s
```

```
              model                        settings    train      val      std
k   p
      Random Forest n_estimators=200, max_depth=5 0.860639 0.755614 0.021807
NaN NaN
                SVM               kernel=rbf, C=1 0.802556 0.746579 0.022317
NaN NaN
Logistic regression                          C=10 0.750489 0.743158 0.028332
NaN NaN
                KNN                     k=43, p=1 0.752707 0.742719 0.026310
43.0 1.0
      Decision Tree                  max_depth=4 0.789211 0.723421 0.026307
NaN NaN
         Perceptron                     eta0=0.1 0.672481 0.663158 0.054060
NaN NaN
```

8. Final evaluation

The tuning in section 7 trained on 70% of the data in each split and held 30% back to measure with. That measuring is now finished, so the final model is trained on all 760 cleaned rows: more data gives a better model, and there is nothing left to hold out for.

The random forest needs no scaling. A tree splits one feature at a time on a threshold, so rescaling a feature cannot change which split is best. That removes the scaler from the final pipeline entirely, and with it the most common way of leaking information into a test set.

In [24]:
```
best_model = RandomForestClassifier(
    n_estimators=200, max_depth=5, random_state=1, n_jobs=-1
)

X_full = data[features].to_numpy()        # all 760 cleaned rows
X_test = test[features].to_numpy()        # same 11 columns, same order, unt

best_model.fit(X_full, y)

print("trained on", X_full.shape[0], "rows x", X_full.shape[1], "features")
print("accuracy on the full training set:", round(best_model.score(X_full, y
print("test matrix ready:", X_test.shape)

importance = (pd.Series(best_model.feature_importances_, index=features)
                .sort_values(ascending=False).round(3))
print("\nwhich features the forest actually used:")
```

```
print(importance.to_string())

# Train our best classifier on the full cleaned training dataset
final_model = RandomForestClassifier(
    n_estimators=200,
    max_depth=5,
    random_state=1
)

final_model.fit(X, y)

# Predict the Kaggle test data
X_test = test[features].to_numpy()
predictions = final_model.predict(X_test)

# Use Kaggle's sample submission to preserve the required format
submission = sample.copy()
submission.iloc[:, -1] = predictions

# Save the submission file
submission.to_csv("submission.csv", index=False)

print(submission.head())
print("Submission shape:", submission.shape)
```

```
trained on 760 rows x 11 features
accuracy on the full training set: 0.8355
test matrix ready: (209, 11)

which features the forest actually used:
alcohol                 0.261
sulphates               0.189
total sulfur dioxide    0.117
volatile acidity        0.106
density                 0.087
chlorides               0.064
pH                      0.044
fixed acidity           0.043
citric acid             0.033
free sulfur dioxide     0.030
residual sugar          0.027
   ID  quality
0   0        1
1   1        1
2   2        0
3   3        0
4   4        0
Submission shape: (209, 2)
```

9. Kaggle submission

The submission uses a column named `quality`, with the DataFrame index named `Id`, so that the row numbers 0 to 208 are written out as the ID column.

Arriving at this took several attempts, because the sources in the folder disagree with each other. The example code on the competition's Overview page writes `Quality` with the index named `ID`; `sample_submission.csv` uses the header `ID,quality`; and `test.csv` carries its own `Id` column running 1117, 1411 and so on. Every upload was rejected with *"ID column Id not found in submission"* until the index was named `Id` rather than `ID`. The values themselves are the row numbers, as in `sample_submission.csv`, not the identifiers from `test.csv`.

No cleaning is applied to `test.csv`. Every row has to receive a prediction, so removing duplicates or outliers there would leave gaps the submission cannot have. Selecting `test[features]` preserves the original row order, so prediction *i* belongs to row *i*.

In [25]:
```
# Submission format copied from the example code on the competition's Overvi
#     y_test = pd.DataFrame(y_test, columns=["Quality"])
#     y_test.index.name = "Id"
#     y_test[['Quality']].to_csv("submission.csv")
y_pred_kaggle = best_model.predict(X_test)

y_test = pd.DataFrame(y_pred_kaggle, columns=["quality"])
y_test.index.name = "Id"                    # row numbers 0..208 become the
y_test[["quality"]].to_csv("submission.csv")

# checks before uploading
check = pd.read_csv("submission.csv")
sample = pd.read_csv("sample_submission.csv")
print("header:            ", ",".join(check.columns))
print("shape:             ", check.shape, "  (expected (209, 2))")
print("IDs match sample:  ", check["Id"].tolist() == sample["ID"].tolist())
print("missing values:    ", int(check.isna().sum().sum()))
print("predicted classes: ", check["quality"].value_counts().to_dict())
print()
print(check.head().to_string(index=False))
```

```
header:            Id,quality
shape:             (209, 2)   (expected (209, 2))
IDs match sample:  True
missing values:    0
predicted classes:  {1: 105, 0: 104}

 Id  quality
  0        1
  1        1
  2        0
  3        0
  4        0
```